

Table 1: Pass-sum execution time (s, median per iteration; sum of pass medians, not full executable wall time) and speedup split by pass type. End-to-end excludes build timing. When present, the OctTree row includes ColorOctree passes; a separate ColorOctree row reports only those passes. ADT fields = non-recursive fields annotated with `@BENCH adt_fields`; SoA bufs = 1+ non-recursive field slots across all constructors (recursive children are stored in the tag buffer). Speedup $>1\times$ means SoA is faster; **bold** marks $>1.1\times$.

Program	ADT fields	SoA bufs	End-to-end			Fold passes			Map passes		
			AoS (s)	SoA (s)	Speedup	AoS (s)	SoA (s)	Speedup	AoS (s)	SoA (s)	Speedup
OctTree	16	9	0.6708	0.6780	0.99×	0.3685	0.3288	1.12 ×	0.3023	0.3492	0.87×
Compiler	9	8	0.2848	0.2188	1.30 ×	0.1253	0.0390	3.21 ×	0.1595	0.1798	0.89×
DBQuery	15	13	0.2563	0.2535	1.01×	0.1183	0.0873	1.36 ×	0.1380	0.1663	0.83×
DecisionTree	7	6	3.821	3.058	1.25 ×	3.821	3.058	1.25 ×	—	—	—
DomTree	15	14	0.1945	0.1428	1.36 ×	0.1361	0.0640	2.13 ×	0.0584	0.0788	0.74×
KDTree	17	16	0.1765	0.1386	1.27 ×	0.1765	0.1386	1.27 ×	—	—	—
LinearListReduction	11	11	0.0291	0.0050	5.79 ×	0.0291	0.0050	5.79 ×	—	—	—
List	2	2	0.5765	0.6368	0.91×	0.3252	0.2997	1.08×	0.2513	0.3370	0.75×
MonoTree	3	2	0.0629	0.0645	0.98×	0.0171	0.0222	0.77×	0.0458	0.0423	1.08×
ObjectGraph	5	4	0.2339	0.2391	0.98×	0.0910	0.0789	1.15 ×	0.1429	0.1602	0.89×
PiecewiseFunctions	8	7	0.3773	0.3493	1.08×	0.1407	0.0900	1.56 ×	0.2366	0.2593	0.91×
TernaryTree	5	3	0.1134	0.1062	1.07×	0.0294	0.0289	1.02×	0.0839	0.0773	1.09×
Trie	10	9	0.2111	0.1992	1.06×	0.0620	0.0320	1.94 ×	0.1491	0.1672	0.89×
ColorOctree	25	18	0.0881	0.0762	1.16 ×	0.0881	0.0762	1.16 ×	—	—	—

Table 2: Baseline Gibbon comparison (times only). Shown variants: AoS-mut, AoS-imm, SoA-mut. Times are full executable end-to-end wall time per run (s), not pass-sum time. **Bold** marks the fastest time across shown variants.

Program	Am	Ai	Sm
Compiler	3.620	—	2.835
DBQuery	3.358	—	3.204
DecisionTree	42.81	—	34.61
DomTree	2.486	—	2.071
KDTree	2.114	—	2.142
LinearListReduction	0.6345	—	0.4064
List	6.819	—	8.054
MonoTree	0.8829	—	1.004
ObjectGraph	3.164	—	3.124
Oct:sumMass	0.6184	—	1.175
Oct:sumEnergy	0.7539	—	1.238
Oct:countActive	0.7952	—	1.176
Oct:countParticles	0.7462	—	1.153
Oct:barnesHutPotential	0.8036	—	1.413
Oct:fmmPotential	1.482	—	2.035
Oct:scaleEnergy	2.158	—	2.955
Oct:clearFlags	2.187	—	2.949
PiecewiseFunctions	4.762	—	4.341
TernaryTree	1.749	—	1.470
Trie	2.549	—	2.656
ColorOctree	1.354	—	1.558

Table 3: Baseline Gibbon comparison (speedups). Shown: AoS-mut/SoA-mut, AoS-imm/AoS-mut, AoS-imm/SoA-mut. Speedups are computed from full executable end-to-end wall time per run. $>1\times$ means the denominator is faster.

Program	Am/Sm	Ai/Am	Ai/Sm
Compiler	1.28 ×	—	—
DBQuery	1.05×	—	—
DecisionTree	1.24 ×	—	—
DomTree	1.20 ×	—	—
KDTree	0.99×	—	—
LinearListReduction	1.56 ×	—	—
List	0.85×	—	—
MonoTree	0.88×	—	—
ObjectGraph	1.01×	—	—
Oct:sumMass	0.53×	—	—
Oct:sumEnergy	0.61×	—	—
Oct:countActive	0.68×	—	—
Oct:countParticles	0.65×	—	—
Oct:barnesHutPotential	0.57×	—	—
Oct:fmmPotential	0.73×	—	—
Oct:scaleEnergy	0.73×	—	—
Oct:clearFlags	0.74×	—	—
PiecewiseFunctions	1.10×	—	—
TernaryTree	1.19 ×	—	—
Trie	0.96×	—	—
ColorOctree	0.87×	—	—

Table 4: Per-pass performance for **OctTree**, OctTree SoA uses 9 buffers; ColorOctree SoA uses 18 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
barnesHutPotential	F	11/16	31%	0.0407 $\pm$ 0.00002	0.0391 $\pm$ 0.00006	1.04 $\times$
countActive	F	10/16	38%	0.0368 $\pm$ 0.00003	0.0291$\pm$0.00001	1.27$\times$
countParticles	F	8/16	50%	0.0355 $\pm$ 0.00004	0.0269$\pm$0.00017	1.32$\times$
fmmPotential	F	12/16	25%	0.0974 $\pm$ 0.00006	0.0954 $\pm$ 0.00002	1.02 $\times$
sumEnergy	F	12/16	25%	0.0351 $\pm$ 0.00002	0.0329 $\pm$ 0.00010	1.07 $\times$
sumMass	F	10/16	38%	0.0348 $\pm$ 0.00002	0.0291$\pm$0.00004	1.20$\times$
clearFlags	M	15/16	6%	0.1530$\pm$0.00041	0.1740 $\pm$ 0.00016	0.88 $\times$
scaleEnergy	M	16/16	0%	0.1493$\pm$0.00027	0.1752 $\pm$ 0.00021	0.85 $\times$
paletteEntriesQuantized	F	13/25	48%	0.0423 $\pm$ 0.00004	0.0335$\pm$0.00010	1.26$\times$
quantizationErrorProxy	F	10/25	60%	0.0458 $\pm$ 0.00008	0.0428 $\pm$ 0.00002	1.07 $\times$
Total				0.6708	0.6780	0.99 $\times$
Geomean						1.09 $\times$

Table 5: Per-pass performance for **Compiler**, ADT has 9 fields, SoA uses 8 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
blockCountPass	F	2/9	78%	0.0119 $\pm$ 0.00001	0.0019$\pm$0.00000	6.20$\times$
branchStatsPass	F	2/9	78%	0.0115 $\pm$ 0.00001	0.0033$\pm$0.00001	3.48$\times$
castInstCountPass	F	3/9	67%	0.0183 $\pm$ 0.00460	0.0013$\pm$0.00001	13.90$\times$
has cycle	F	4/9	56%	0.0241 $\pm$ 0.00095	0.0176$\pm$0.00132	1.37$\times$
instCountPass	F	2/9	78%	0.0119 $\pm$ 0.00002	0.0019$\pm$0.00001	6.21$\times$
latencyModelPass	F	3/9	67%	0.0118 $\pm$ 0.00001	0.0029$\pm$0.00001	4.12$\times$
memoryOpStatsPass	F	3/9	67%	0.0117 $\pm$ 0.00002	0.0037$\pm$0.00001	3.13$\times$
throughputModelPass	F	3/9	67%	0.0118 $\pm$ 0.00001	0.0029$\pm$0.00000	4.06$\times$
verifyIR	F	9/9	0%	0.0123 $\pm$ 0.00002	0.0034$\pm$0.00001	3.62$\times$
stripSideEffectsPass	M	7/9	22%	0.0798$\pm$0.00051	0.0900 $\pm$ 0.00014	0.89 $\times$
targetReturnPass	M	9/9	0%	0.0797$\pm$0.00054	0.0898 $\pm$ 0.00034	0.89 $\times$
Total				0.2848	0.2188	1.30$\times$
Geomean						3.22$\times$

Table 6: Per-pass performance for **DBQuery**, ADT has 15 fields, SoA uses 13 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
countJoins	F	3/15	80%	0.0185 $\pm$ 0.00002	0.0109$\pm$0.00004	1.70$\times$
filterSelectivitySkew	F	5/15	67%	0.0183 $\pm$ 0.00002	0.0124$\pm$0.00003	1.48$\times$
hashJoinPressure	F	5/15	67%	0.0199 $\pm$ 0.00003	0.0130$\pm$0.00003	1.53$\times$
sumCost	F	6/15	60%	0.0184 $\pm$ 0.00002	0.0149$\pm$0.00004	1.23$\times$
sumMemory	F	6/15	60%	0.0185 $\pm$ 0.00002	0.0145$\pm$0.00007	1.28$\times$
sumRows	F	6/15	60%	0.0246 $\pm$ 0.00012	0.0216$\pm$0.00012	1.14$\times$
clearQueryFlags	M	14/15	7%	0.0694$\pm$0.00808	0.0828 $\pm$ 0.00008	0.84 $\times$
scaleCosts	M	15/15	0%	0.0686$\pm$0.00022	0.0835 $\pm$ 0.00008	0.82 $\times$
Total				0.2563	0.2535	1.01 $\times$
Geomean						1.22$\times$

Table 7: Per-pass performance for **DecisionTree**, ADT has 7 fields, SoA uses 6 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
classify Batch	F	5/7	29%	3.279 $\pm$ 0.0305	2.712$\pm$0.0241	1.21$\times$
classify Depth	F	4/7	43%	0.0469 $\pm$ 0.00002	0.0369$\pm$0.00009	1.27$\times$
classify tree	F	5/7	29%	0.0469 $\pm$ 0.00002	0.0370$\pm$0.00786	1.27$\times$
countFeatureUses	F	3/7	57%	0.0501 $\pm$ 0.00002	0.0324$\pm$0.00005	1.55$\times$
countLeaves	F	2/7	71%	0.0475 $\pm$ 0.00009	0.0248$\pm$0.00004	1.91$\times$
countNodes	F	2/7	71%	0.0475 $\pm$ 0.00003	0.0263$\pm$0.00007	1.80$\times$
countSmallLeaves	F	3/7	57%	0.0510 $\pm$ 0.00002	0.0322$\pm$0.00007	1.58$\times$
inferenceCost	F	2/7	71%	0.0504 $\pm$ 0.00002	0.0290$\pm$0.00004	1.73$\times$
sumImpurity	F	3/7	57%	0.0501 $\pm$ 0.00003	0.0326$\pm$0.00122	1.54$\times$
sumPathLengths	F	3/7	57%	0.0524 $\pm$ 0.00002	0.0340$\pm$0.00004	1.54$\times$
sumSamples	F	3/7	57%	0.0482 $\pm$ 0.00170	0.0321$\pm$0.00005	1.50$\times$
treeDepth	F	2/7	71%	0.0509 $\pm$ 0.00004	0.0281$\pm$0.00002	1.81$\times$
Total				3.821	3.058	1.25$\times$
Geomean						1.54$\times$

Table 8: Per-pass performance for **DomTree**, ADT has 15 fields, SoA uses 14 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
count styled	F	3/15	80%	0.0337 $\pm$ 0.00002	0.0115$\pm$0.00004	2.94$\times$
find max Bottom	F	5/15	67%	0.0337 $\pm$ 0.00002	0.0226$\pm$0.00006	1.49$\times$
SumArea	F	6/15	60%	0.0349 $\pm$ 0.00003	0.0183$\pm$0.00012	1.91$\times$
sumTextWidth	F	3/15	80%	0.0337 $\pm$ 0.00003	0.0116$\pm$0.00005	2.90$\times$
computeWidths	M	14/15	7%	0.0296$\pm$0.00008	0.0442 $\pm$ 0.00012	0.67 $\times$
scaleLayout	M	15/15	0%	0.0288$\pm$0.00011	0.0346 $\pm$ 0.00010	0.83 $\times$
Total				0.1945	0.1428	1.36$\times$
Geomean						1.54$\times$

Table 16: Per-pass performance for **Trie**, ADT has 10 fields, SoA uses 9 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across `-iterate` runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
autocompleteTopKProxy	F	5/10	50%	0.0139 $\pm$ 0.00002	0.0080$\pm$0.00009	1.74 $\times$
countLazyNodes	F	4/10	60%	0.0125 $\pm$ 0.00003	0.0067$\pm$0.00002	1.87 $\times$
countTerminals	F	3/10	70%	0.0118 $\pm$ 0.00001	0.0058$\pm$0.00004	2.03 $\times$
sumPrefixFreq	F	3/10	70%	0.0119 $\pm$ 0.00002	0.0058$\pm$0.00002	2.07 $\times$
sumSubtreeHints	F	3/10	70%	0.0119 $\pm$ 0.00001	0.0058$\pm$0.00002	2.06 $\times$
decayTrieStats	M	10/10	0%	0.0745$\pm$0.00039	0.0846 $\pm$ 0.00035	0.88 $\times$
resetTraversalState	M	10/10	0%	0.0746 $\pm$ 0.00038	0.0825 $\pm$ 0.00016	0.90 $\times$
Total				0.2111	0.1992	1.06 $\times$
Geomean						1.56 $\times$